CMPT 365 Multimedia Systems

Media Compression
- Audio

Fall 2023

Approximate file sizes for **1 second** of audio

Channels	Resolution #Bits Quantization	Fs #sample/s	File Size
Mono	8bit	8Khz	64Kb Already 64Kbs
Stereo	8bit	8Khz	128Kb
Mono	16bit	8Khz	128Kb
Stereo	16bit	16Khz	256Kb
Stereo	16bit	44.1Khz	1441Kb*
Stereo	24bit	44.1Khz	2116Kb

1CD 700M 70-80 mins Audio

Therefores needA encode

Outline

- Lossless Audio Coding
- Lossy Audio Coding
- MPEG audio (MP3)

Lossless coding

No information loss
↳ perfectly recover data

- Algorithm lossless cause recovers value Prediction Method fixed
  - Linear prediction: estimate what the value a sample will have, based on previous samples imperfect

    previous 2
    ↑
    n = 5 : p = 2
    Means value n = 3 n = 4
    Predict n = 5

    n sample Prediction

$$x'(n) = \sum_{i=1}^{p} a_i x(n-i)$$

    ↳ previous sample

    ↳ encode difference
    ↳ should b small if prediction GREAT
    ↳ ex n = 5 = 10 (Actual) } 2 (only encode 2)
    x'(5) = 8
    Prediction + Error = Original
    x'(5) + 2 = 10 lossless

  - Channel coupling - mid/side-coding:
    - calculates a "mid"-channel by addition of left and right channel (l+r)/2 and a "side"-channel (l-r)/2. Where l - left channel sample value
      r - right channel sample value
    Only needs 2 channels
    Arithmetic gives original
  - Entropy coding

    includes Huffman, LZW, Arithmetic can b used.

Lossless coding

- **FLAC (Free Lossless Audio Codec)**
  - Golomb-Rice coding for entropy coding
    - closely related to Huffman/Arithmetic

- **Monkey's Audio – APE format**
  - Range coding for entropy coding
    - Similar to Arithmetic Coding
  - Slightly better than FLAC (but time-consuming decoding)
  - Officially only for Windows

Performance of Lossless Coding

Affects Realtime Performance

- http://members.home.nl/w.speek/comparison.htm

Important →

Compressor	Options	Ratio*	Encoding Speed**	Decoding Speed**
		%	x realtime	x realtime
La 0.4b	default	55.5	2.1	2.7
OptimFROG 4.509	highnew	55.8	1.1	1.5
Monkey's Audio 3.99	extra high	56.4	8.8	8.7
OptimFROG 4.509	default	56.7	6.0	8.9
Monkey's Audio 3.99	high	56.9	15.8	14.3
Monkey's Audio 3.99	normal	57.3	18.1	16.0
WavPack 4.0	high	58.0	16.0	16.1
WMA 9	default	58.0	9.4	10.8
RKAU 1.07	fast (l1)	58.4	8.0	9.6
LPAC Archiver 1.41	medium, JS, random access	58.8	13.5	24.6
LPAC Archiver 1.41	extra high, JS, random access	58.8	8.1	20.2
TTA 3.2	default	59.0	26.6	23.4
WavPack 4.0	normal	59.4	26.3	28.4
FLAC 1.1.2	8	59.6	4.2	44.7
FLAC 1.1.2	default (5)	59.8	19.9	44.7
Apple Lossless (iTunes 4.7)	automatic	60.0	15.6	35.0
Shorten 3.6.0	default	63.7	33.7	70.9

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- Lossless Audio Coding
- Lossy Audio Coding
- MPEG audio (MP3)

Lossy coding: Perceptual Coding

Cannot hear/see no needA encode

- Hide errors where humans will not see or hear it
  - Study hearing and vision system to understand how we see/hear ⇒ Understand what's interpretable
  - Masking refers to one signal overwhelming/hiding another (e.g., loud siren or bright flash)
    ↳ loud noise, cannot hear quiet ↳ Bright light cover other light
- Natural Bandlimitng
  - Audio perception is 20-20 kHz but most sounds in low frequencies (e.g., 2 kHz to 4 kHz) no needA encode every frequecy only lower
  - Low frequencies may be encoded as single channel cause poor Determine Direction
  - Human ear can tolerate 200ps delay

    if 2 sound within 200ps delay cAnnot Distinguish

    ↳ encode together compresses DAPA

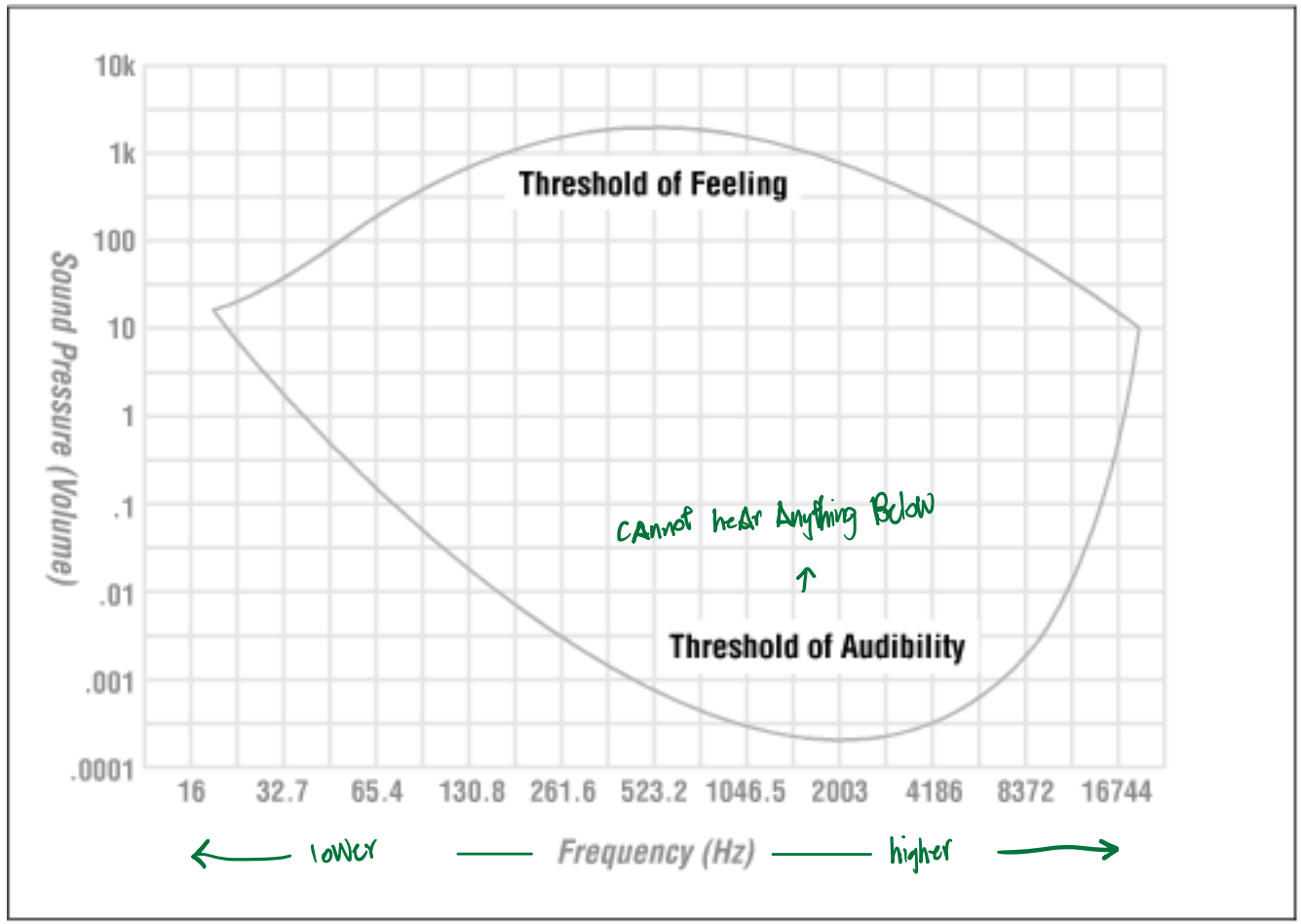
These lossy cause cAnnot recovers

Psychoacoustic

-Human aural response

Human Audibility

loudness



cannot hear anything below

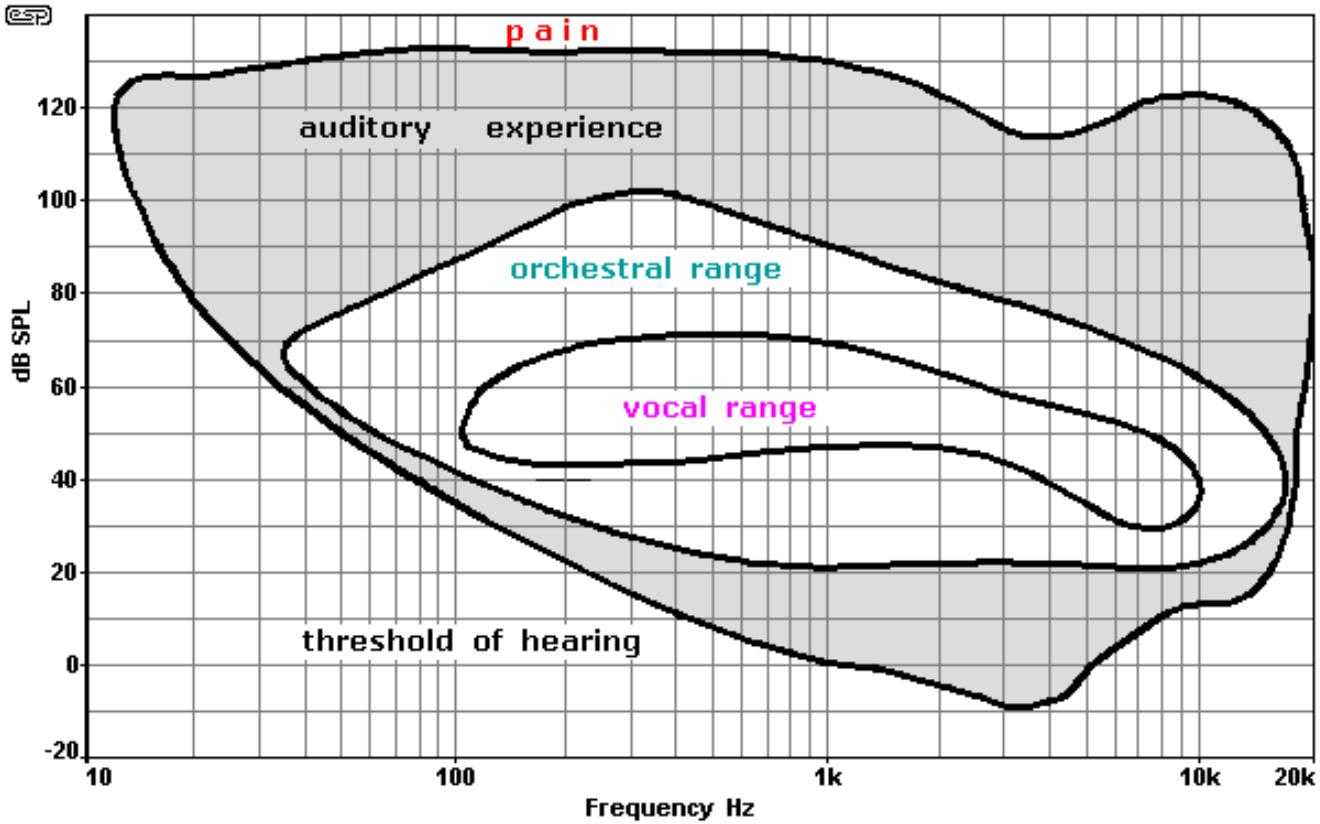
lower — higher

Ends needs B loud to Audible

Psychoacoustic

-Human aural response

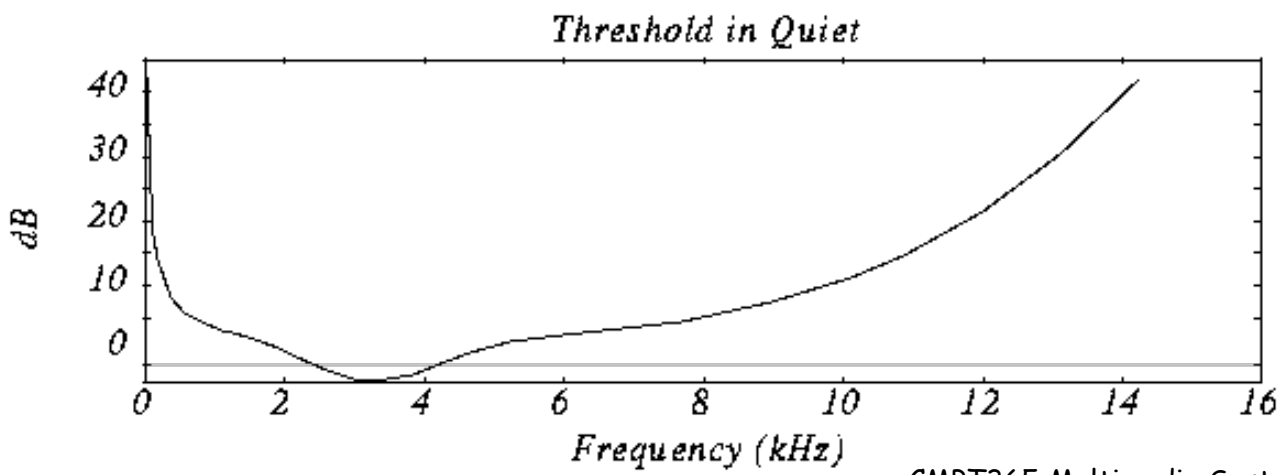
Shows encoding thresholds



Frequency Maskings

Psychoacoustic Model

- Basically: If you can't hear the sound, don't encode it
  - Frequency range is about 20 Hz to 20 kHz, most sensitive at 2 to 4 KHz.
  - Dynamic range (quietest to loudest) is about 96 dB → 16 Bits
  - Normal voice range is about 500 Hz to 2 kHz
    - Low frequencies are vowels and bass
    - High frequencies are consonants
- Sensitivity experiment
  - Experiment: Put a person in a quiet room. Raise level of 1 kHz tone until just barely audible. Vary the frequency and plot



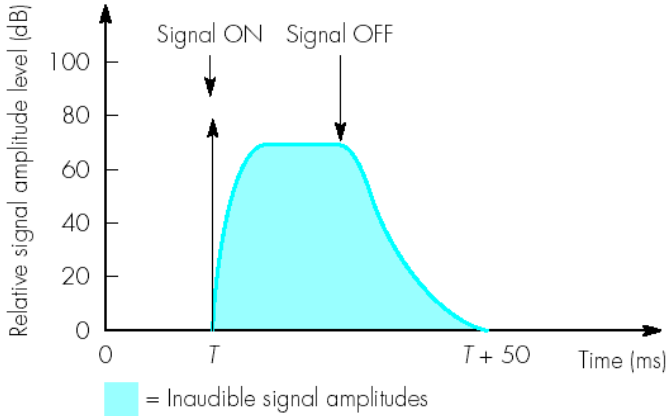
time occurence

Psychoacoustic Model con'td

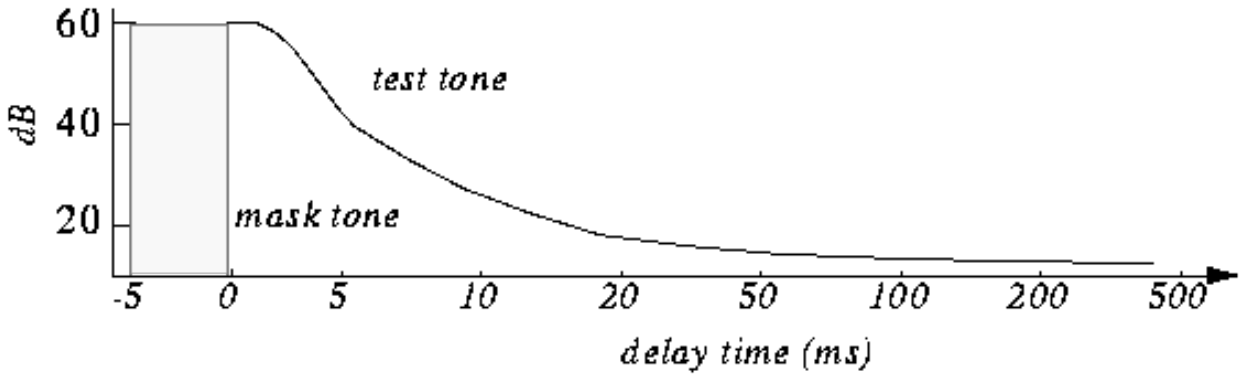
- Temporal masking: If we hear a loud sound, it takes a little while until we can hear a soft tone nearby.

loud → quiet
↳ cannot hear quiet cause mask
quiet → loud
↳ cannot hear

TEMPORAL Apply Before & After



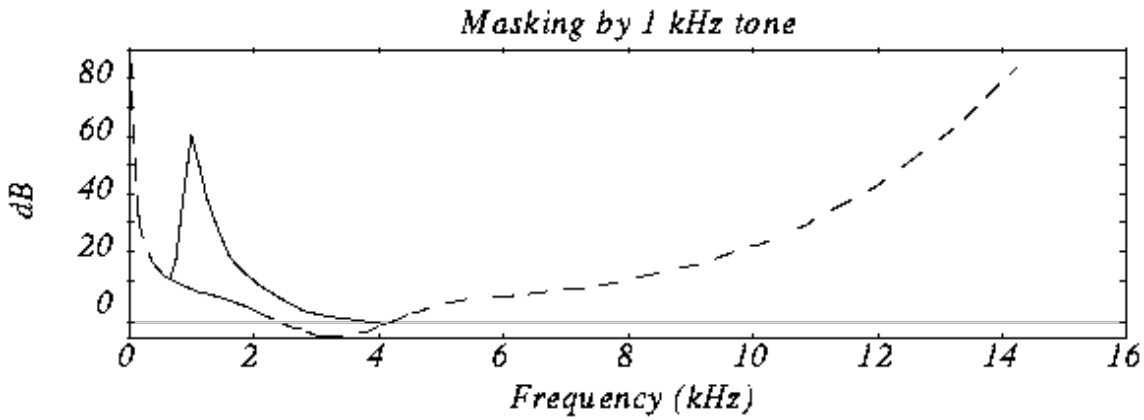
- Experiment:
  - Play 1 kHz *masking tone* at 60 dB, plus a *test tone* at 1.1 kHz at 40 dB. Test tone can't be heard (it's masked). Stop masking tone, then stop test tone after a short delay.
  - Adjust delay to the shortest time when test tone can be heard.
  - Repeat with different level of the test tone and plot:

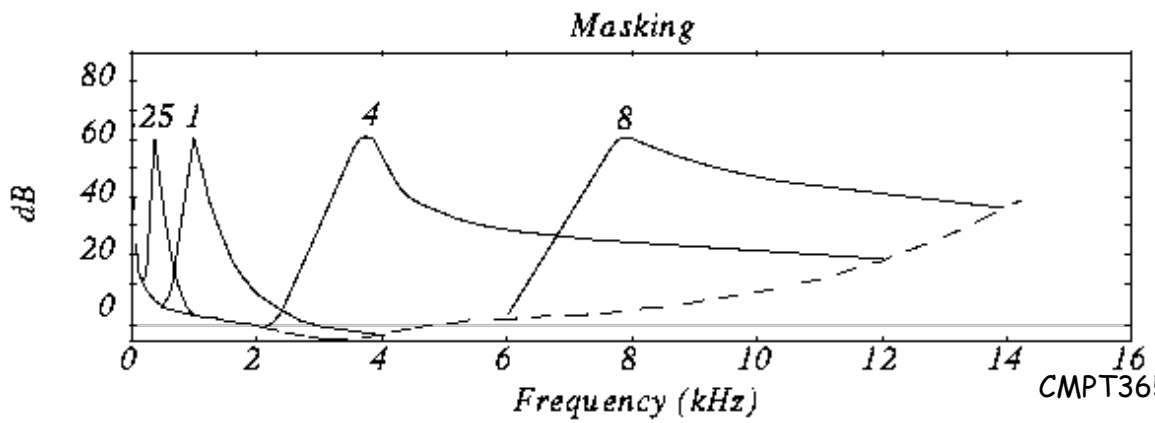


Psychoacoustic Model con'td

- Frequency masking: Do receptors interfere with each other?
- Experiment:
  - Play 1 kHz tone (*masking tone*) at fixed level (60 dB). Play *test tone* at a different level and raise level until just distinguishable.
  - Vary the frequency of the test tone and plot the threshold when it becomes audible:

Similar sound

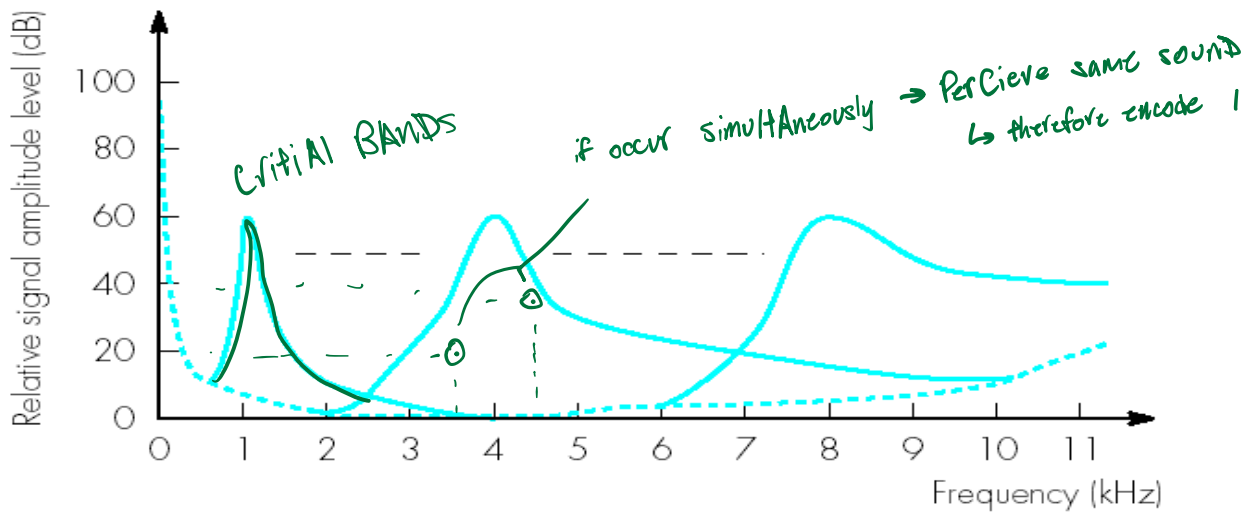




Psychoacoustic Model con'td

if 2 sound occur simultaneously within same band encode 1

- Frequency masking: If within a critical band a stronger sound and weaker sound compete, you can't hear the weaker sound. Don't encode it.

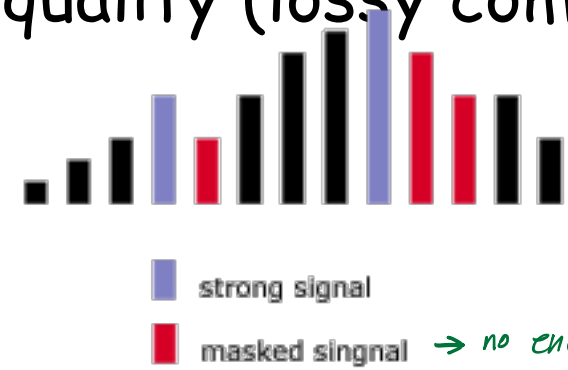


Our brains perceive the sounds through 25 distinct ***critical bands***. The bandwidth grows with frequency (above 500Hz).

- At 100Hz, the bandwidth is about 160Hz;
- At 10kHz it is about 2.5kHz in width.

Perceptual Coding

- Makes use of psychoacoustic knowledge to reduce the amount of information required to achieve the same perceived quality (lossy compression)



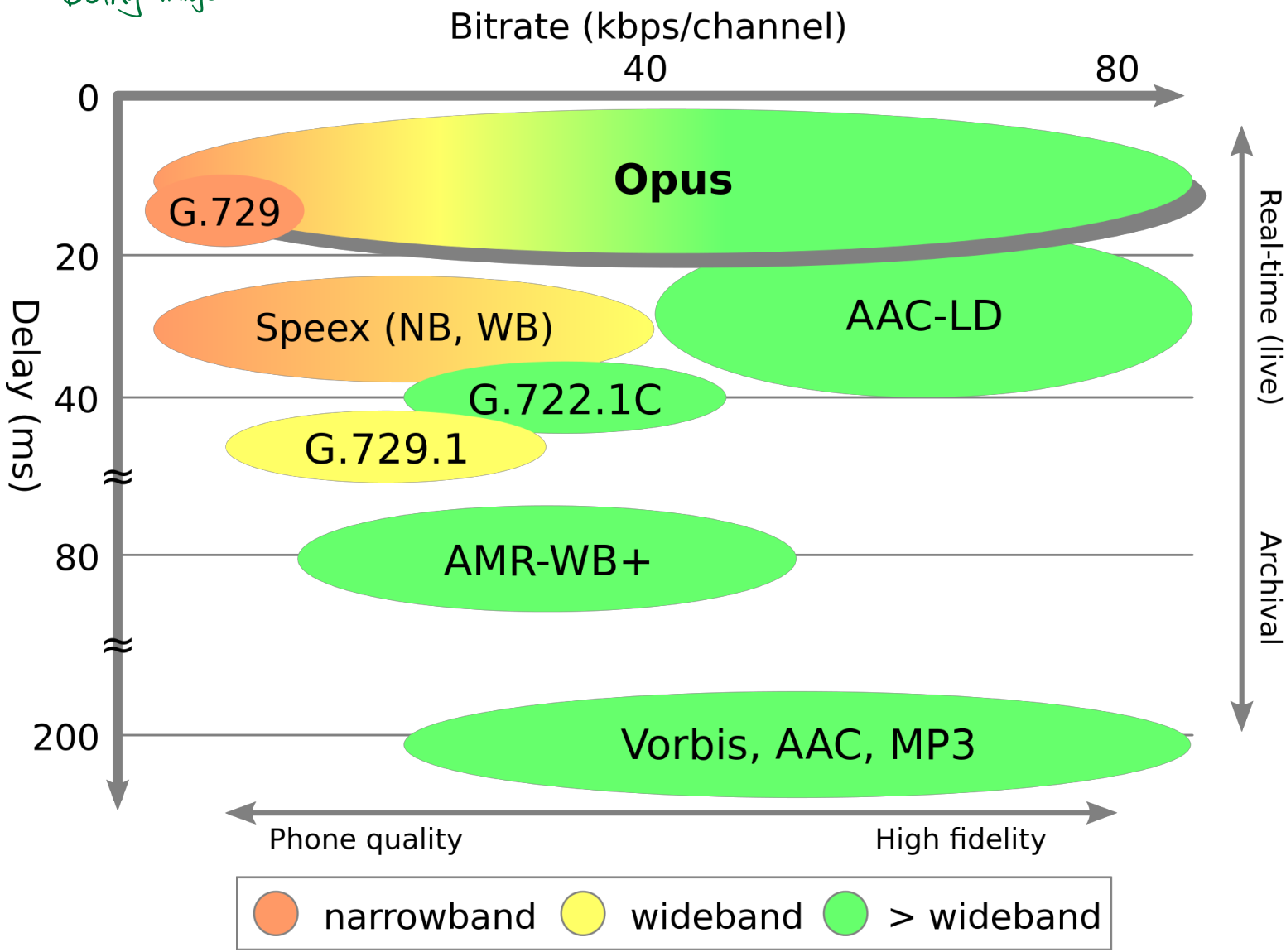
→ no encoding necessary
↳ cannot hear

- Example:
  - Sony MiniDisc uses Adaptive TRAnsform Coding (ATRAC) to achieve a 5:1 compression ratio (about 141 kbps)
  - MPEG audio (MP3)

http://www.mpeg.org
http://www.minidisc.org/aes_atrac.html

Perceptual Coding

Delay large → ≠ Realtime



Outline

- Lossless Audio Coding
- Lossy Audio Coding
- MPEG audio (MP3)

MPEG (Moving Picture Expert Group) Audio

MP 3 is MPEG1 Layer

- MPEG-1: 1.5 Mbits/sec for audio and video
  - About 1.2 Mbits/sec for video, 0.3 Mbits/sec for audio
  - Cf. Uncompressed CD audio is 44,100 samples/sec * 16 bits/sample * 2 channels > 1.4 Mbits/sec need a compress 1.4 to 0.3
  - Compression factor ranging from 2.7 to 24.

- With Compression rate 6:1 (16 bits stereo sampled at 48 KHz is reduced to 256 kbits/sec), expert could not distinguish

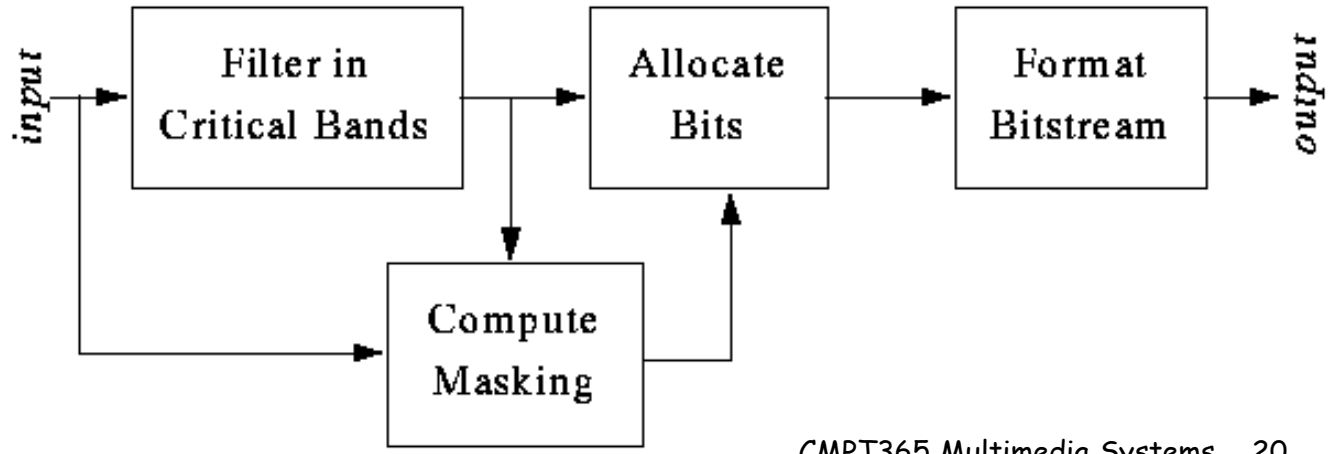
MPEG Audio Layers

- MPEG defines 3 layers for audio. Basic model is same, but codec complexity increases with each layer.
- Layer 1: DCT type filter with one frame and equal frequency spread per band. Psychoacoustic model only uses frequency masking.
- Layer 2: Three frames in filter (before, current, next, a total of 1152 samples). This models a little bit of the temporal masking. Before & After → Temporal
- Layer 3 (MP3): Better critical band filter is used (non-equal frequencies), psychoacoustic model includes temporal masking effects, takes into account stereo redundancy, and uses Huffman coder. PsychAcoustic
- **Stereo Redundancy Coding:**
  - Intensity stereo coding -- at upper-frequency subbands, encode summed signals instead of independent signals from left and right channels.
  - Middle/Side (MS) stereo coding -- encode middle (sum of left and right) and side (difference of left and right) channels.

Algorithm

- Divide the audio signal (e.g., 48 kHz sound) into 32 frequency subbands --> *subband filtering*.
  - Modified discrete cosine transform (MDCT) -
- Masking for each band caused by nearby band

  Check masks
  ↳ if 1 BAND cover Another
  ↳ encode 1
  - *psychoacoustic model*
  - If the power in a band is below the masking threshold, don't encode it.
  - Otherwise, determine number of bits needed to represent the coefficient such that noise introduced by quantization is below the masking effect
    - One fewer bit introduces about 6 dB of noise).
- Format bitstream



Example

- After analysis, the first levels of 16 of the 32 bands:

Band	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Level (db)	0	8	12	10	6	2	10	60	35	20	15	2	3	5	3	1

→ masks 12

- If the level of the 8th band is 60dB, it gives a masking of 12 dB in the 7th band, 15dB in the 9th.
- Level in 7th band is 10 dB (< 12 dB), so ignore it.
- Level in 9th band is 35 dB (> 15 dB), so send it. Only encode 35 - 15 = 20 DB

[Only the amount above the masking level needs to be sent, so instead of using 6 bits to encode it, we can use 4 bits -- a saving of 2 bits (= 12 dB).]

QUANT SQNR 6.02 × N
6.02 × 4 = 24 cover 20 so 4 bits enough

Effectiveness of MPEG Audio

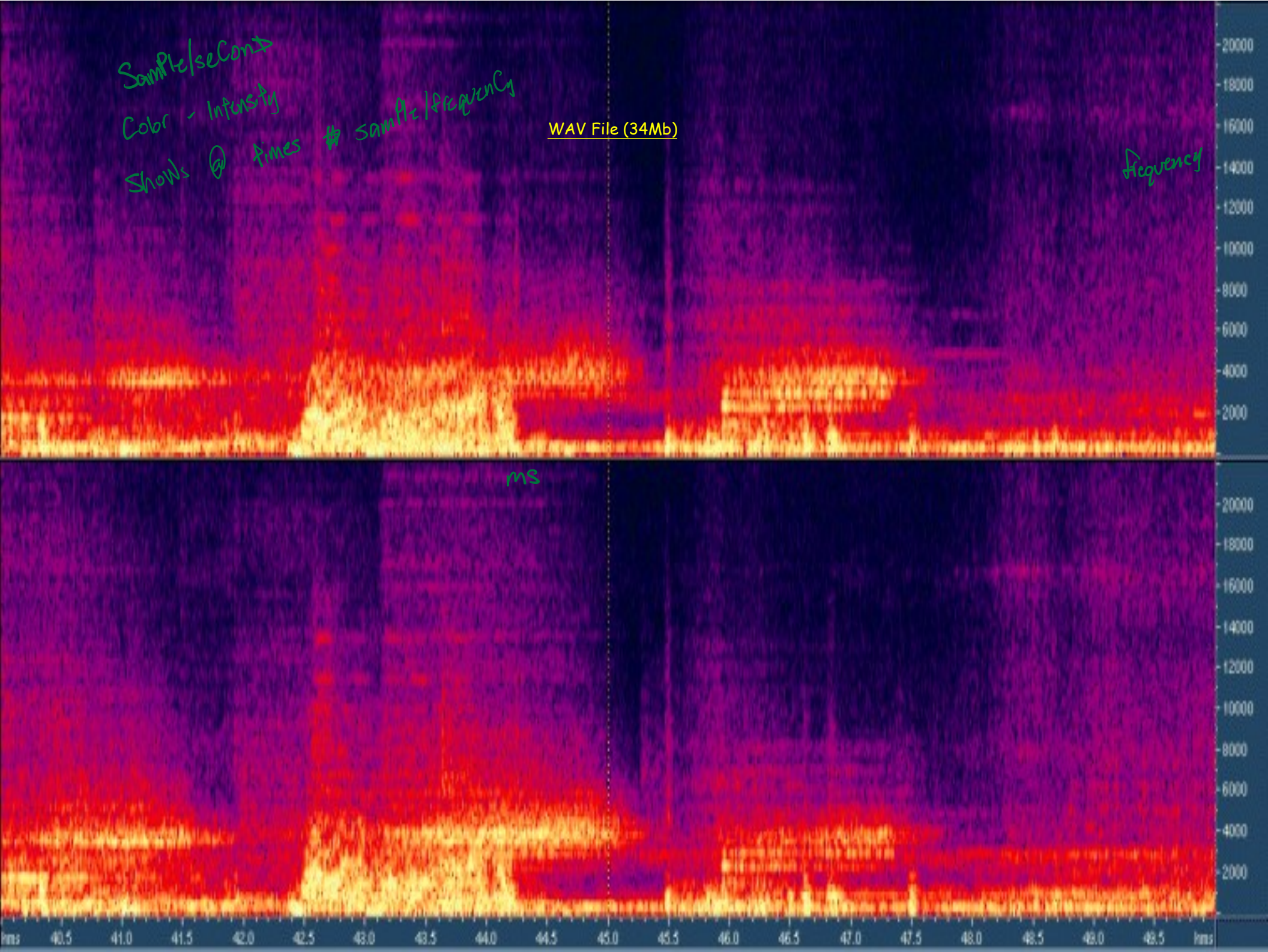
Layer	Target Bit-rate	Ratio	Quality at 64 kb/s	Quality at 128 kb/s	Theoretical Min. Delay
Layer 1	192 kb/s	4:1	---	---	19 ms
Layer 2	128 kb/s	6:1	2.1 to 2.6	4+	35 ms
Layer 3	64 kb/s	12:1	3.6 to 3.8	4+	59 ms

GREAT only takes long.

- **Quality factor**: 5 - perfect, 4 - just noticeable, 3 - slightly annoying, 2 - annoying, 1 - very annoying
- **Real delay** is about 3 times of the theoretical delay

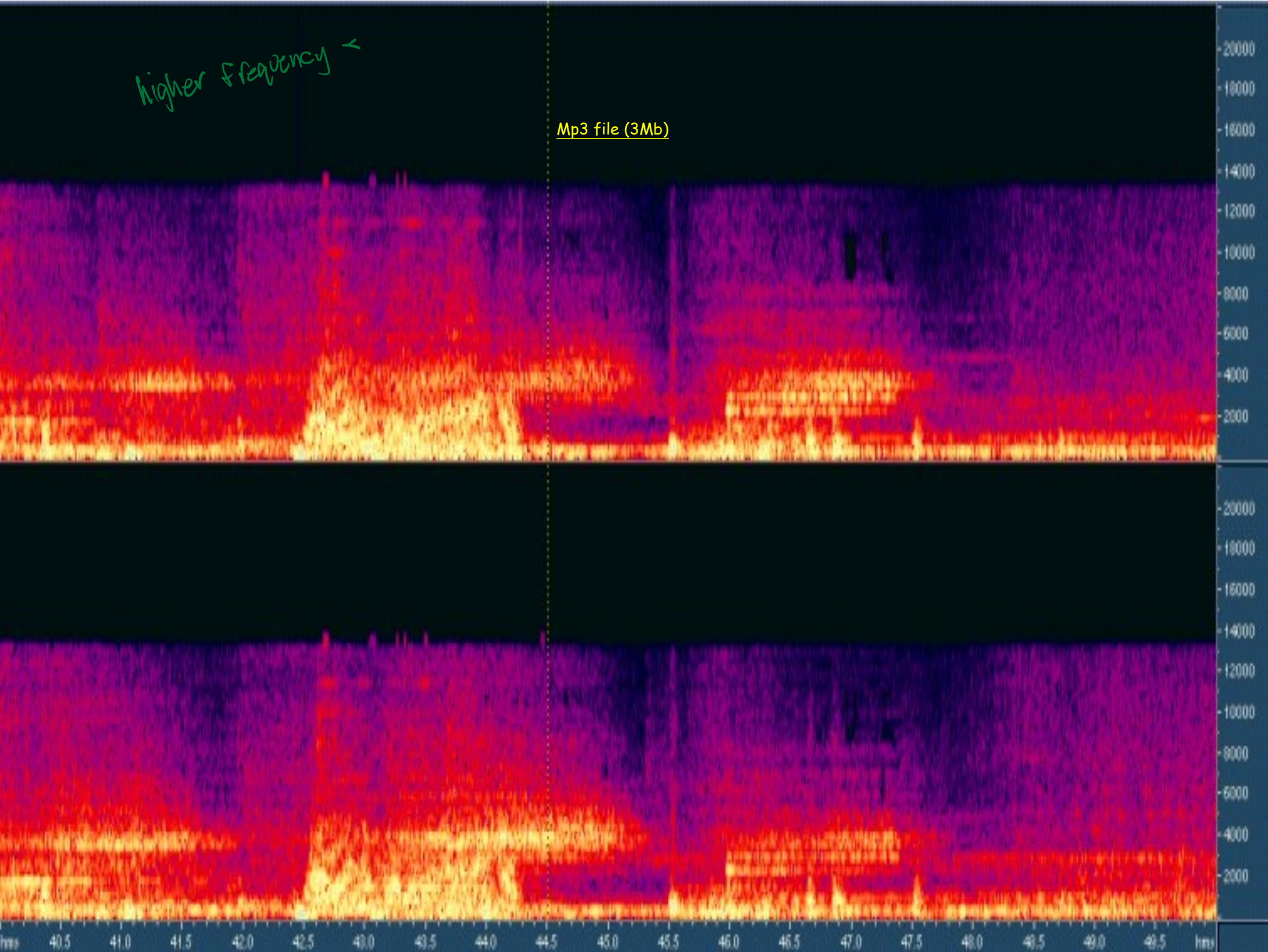
Artefacts of compression

- Mp3 encoded recordings rarely sound identical to original uncompressed audio files
- Whole areas of the spectrum are lost in the encoding process
- On small domestic 'hi-fi' or PC speakers, however, mp3 compressed audio can be acceptable


WAV File (34Mb)
Sample/seconds
Color - Intensity
Shows @ times # sample/frequency
Frequency
ms
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
ms
40.5
41.0
41.5
42.0
42.5
43.0
43.5
44.0
44.5
45.0
45.5
46.0
46.5
47.0
47.5
48.0
48.5
49.0
49.5
ms

higher frequency <

Mp3 file (3Mb)



Artefacts of compression

- Sound Test: Difference between WAV and MP3
  - http://www.noiseaddicts.com/2010/04/sound-test-difference-between-wav-vs-mp3/
- Do 320kbps mp3 files really sound better (vs 128kbps) for your ear?
  - www.noiseaddicts.com/2009/03/mp3-sound-quality-test-128-320/

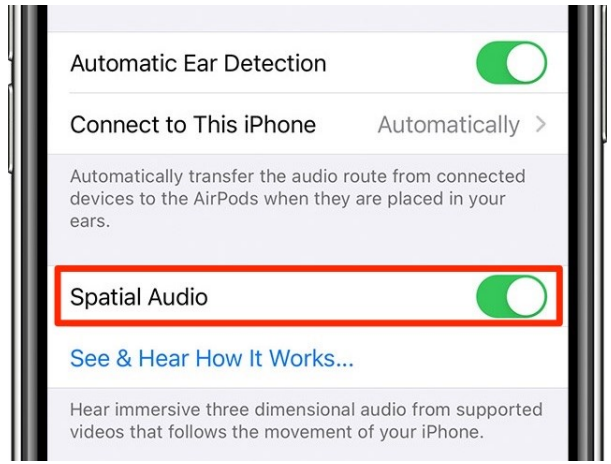
Advanced Audio Coding (AAC)

- An improvement to MP3
  - Part of MPEG-2/4 specifications
  - Standard format for iTunes, DivX, Playstation 3, Android, Blackberry ...
- More sample frequencies (8-96 KHz)
  - 16-48 for MP3
  - Better handling of frequencies above 16 kHz
- Up to 48 channels
  - 2 channel in original MP3 and 5.1 channel in current MP3
- Higher efficiency and accuracy
  - Additional modules/tools to increase compression efficiency

Future of Digital Audio

- Ultra high compression
  - Internet streaming
  - Internet phone
- High Definition audio
  - CD (compact disc - Redbook)
    - Sony 1976; Philips 1979
  - SACD (super audio CD)
    - Sony/Philips 1999
  - DVD-A (DVD Audio)
    - Toshiba 2000
    - 8.5GB disc
    - A war with SACD (HDDVD/BluRay later)
    - Neither successful in consumer market
- Spatial audio

	CD	SACD
Format	16 bit PCM	1 bit DSD
Sampling frequency	44.1 kHz	2.8224 MHz
Dynamic range	96 dB	120 dB
Frequency range	20 Hz - 22.05 kHz	20 Hz - 50 kHz
Disc capacity	700 MB	7.95 GB
Stereo	Yes	Yes



AlreADy GREAT